

Cord Prolapse and Presentation Guideline

1. Scope

Site	Operational Area	Applicable to
Armada Kalamunda Group	Obstetrics and Gynaecology	Medical and Midwifery

2. Purpose

To provide guidance for the prevention, diagnosis and management of cord prolapsed.

3. Background

Incidence of cord prolapse ranges from 0.1% to 0.6%. In breech presentation the incidence is 1% cases involving cord prolapse consistently appear in the perinatal mortality enquiries.

Neonatal asphyxia in these circumstances is due to:

- Cord compression (preventing venous return to the fetus)
- Umbilical vasospasm (preventing venous and arterial blood flow to and from the fetus) due to exposure to external environment.

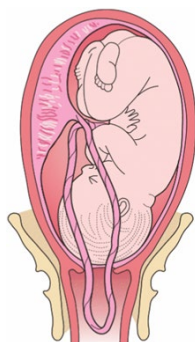


Fig 1: Cord prolapse

4. Procedure

Risk factors associated with cord prolapsed

Table 1: Associations with cord prolapse (and cord presentation)

General	Procedure-related
Multiparity	Artificial rupture of membranes with high presenting part
Low birthweight (< 2.5 kg)	Vaginal manipulation of the fetus with ruptured membranes
Preterm labour (< 37+0 weeks)	External cephalic version (during procedure)
Fetal congenital anomalies	Internal podalic version
Breech presentation	Stabilising induction of labour

General	Procedure-related
Transverse, oblique and unstable lie*	Insertion of intrauterine pressure transducer
Second twin	Large balloon catheter (e.g. Cook) induction of labour
Polyhydramnios	
Unengaged presenting part	
Low-lying placenta	

*Unstable lie is when the longitudinal axis of the fetus (lie) is changing repeatedly after 37+0 weeks.

4.1 Preventive measures

- Elective admission to hospital after 37+0 weeks of gestation should be discussed with women with transverse, oblique or unstable lie, and women in the community should be advised to present urgently if there are signs of labour or suspicion of membrane rupture.
- Inpatient care should be recommended for women with non-cephalic presentations and preterm pre-labour rupture of membranes.
- Whenever possible avoid artificial rupture of membrane (ARM) if the presenting part is mobile and/or high.
- If it becomes unavoidable to rupture the membranes with a high presenting part, this should be performed with arrangements in place for immediate caesarean section.
- Keep to a minimum upward pressure on the presenting part during vaginal examination and other obstetric interventions in the context of ruptured membranes because of the risk of upward displacement of the presenting part and cord prolapse.
- Avoid ARM if cord is digitally palpated on vaginal examination below the presenting part.
- NB: When cord presentation is diagnosed in established labour, caesarean section is usually indicated.

4.2 Diagnosis of umbilical cord prolapse or presentation

Diagnosis of cord prolapse is made by visual inspection via speculum examination, at the introitus or by palpation during digital vaginal examination where the umbilical cord is felt below or beside the presenting part.

Cord prolapse should be suspected with an abnormal fetal heart rate pattern (bradycardia, severe variable decelerations) occurring soon after spontaneous or ARM.

The cord may be extruding from the vagina or wrapped around the presenting part.

Note: In the presence of predisposing risk factors a vaginal examination should always be performed after the membranes rupture spontaneously or by ARM or if a fetal bradycardia occurs after rupture of membranes.

Speculum or digital vaginal examination should be performed at preterm gestation when cord prolapse is suspected.

5. Management

5.1 Cord presentation

- If cord presentation is found, immediately inform the GP Obstetrician(GPO) and Specialist Obstetrician.
- Commence continuous CTG and prepare for birth.

5.2 Cord prolapse

- Press the staff assist button, dial 55 and call a Code blue CS to theatre to alert GPO, Specialist Obstetrician, Anaesthetist and Paediatrician on call. If after hours follow the same process and inform Specialist Obstetrician and Paediatrician immediately via mobile phone.
- Initiate immediate assessment of clinical circumstances; gestation, presentation, cervical dilatation, fetal wellbeing.
- Immediate delivery is necessary when the fetus is viable.

5.3 Discontinue intravenous oxytocin (if being administered)

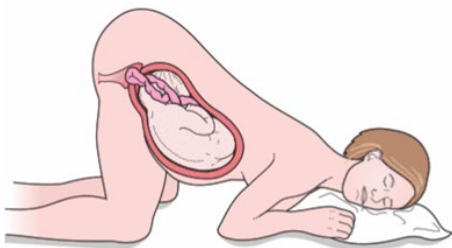
Options to relieve cord compression:

- Manual elevation of the presenting part by inserting a gloved hand in the vagina and pushing it upwards and above the pelvic brim with woman in exaggerated SIMS position (left lateral supported with 2 pillows).
- Lower head of bed.



Exaggerated SIMS position

- All fours with head below level of bottom is an alternative position(knee chest position)



Knee chest position

- **Then:** Insert a Foleys catheter and rapidly fill the bladder with 500 – 700mls Normal Saline at room temperature to elevate the fetal presenting part if a delay in expediting birth is anticipated. Clamp the catheter. The clamp must be released, and the bladder drained before any attempt to deliver the baby.
- Oxygen via mask at 8L/min

- To prevent vasospasm of the cord there should be minimum handling of loops of cord lying outside the vagina.
- Apply a dry pad to try to keep cord inside vagina
- Tocolysis can be considered to reduce contractions while preparing for caesarean section. The suggested tocolytic regimen is Terbutaline 0.25mg subcutaneously.
- Continuous fetal monitoring should be maintained, and the maternal pulse palpated to differentiate. If fetal viability is in doubt, an ultrasound scan should be performed.
- If fetal death in utero is diagnosed, vaginal birth can be considered.

5.4 Delivery

- Assess and assist birth by the most appropriate means.
- Urgency of birth is dependent on Fetal heart rate.
- Category 1 caesarean section is indicated if birth is not imminent, the fetal heart rate pattern is pathological or there is fetal bradycardia. Call a Code Blue CS to Theatre and Code Blue Neonates. Obtain consent, which may need to be verbal in a category 1 situation.
- Simultaneously site IV cannula and obtain bloods for FBP and cross match.
- Provide reassurance and explanation to the woman and support partner.
- Assisted vaginal birth can be considered if conditions are suitable (e.g. the woman is fully dilated and the presenting part is at spines or below and is a multigravida)
- Document the time care given and times of decisions made.
- Obtain paired umbilical cord blood samples for pH and metabolic analysis.
- Arrange for a full debriefing about the whole incident and document the conversation.
- Debrief staff involved.
- An appointment to Specialist Clinic should be made to discuss the next pregnancy and birth.

6. Cord prolapse in the community setting

- Midwives should assess the risk of cord prolapse for women requesting home birth and at the start of labour in the community.
- Women with known cord prolapse should be advised by telephone to assume the knee–chest face-down position while waiting for emergency ambulance transfer to hospital.
- During emergency ambulance transfer, the knee–chest position is potentially unsafe and the exaggerated Sims position (left lateral with pillow under hip) should be used.
- All women with cord prolapse should be advised to be transferred to the nearest hospital with caesarean section facility, unless an immediate vaginal examination by a competent professional reveals that a spontaneous vaginal birth is imminent.
- The presenting part should be elevated during transfer either manually or by using bladder distension.
- To prevent vasospasm, there should be minimal handling of loops of cord lying outside the vagina. Use of dry pad/ gauze is recommended.

7. Legislative context

The following documents are required to give effect to this procedure:

- Health Department of Western Australia (HDWA) [Clinical Handover Policy MP0095](#)

- East Metropolitan Health Service (EMHS) [Inter-Hospital Patient Transfer Policy EMHS: 38](#)
- EMHS [Patient Identification and Procedure Matching EMHS: 21](#)

8. Supporting information

The following documents support the implementation of this procedure:

- [Clinical Handover Procedure AKG-PRO-0521-19](#)
- [Ambulance Transfer of an Obstetric Adult to a Tertiary Hospital Procedure OBS-PRO-0031-13](#)
- [Escalation – Clinical Conflict Policy OBS-GUI-0160-15](#)
- [Recognition and Response to Acute Deterioration Policy AKG-POL-0388-14](#)
- EMHS [Appendix A: IHPT Roles, Responsibilities and Escalation](#)

9. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Specific findings are to be reported at the departmental meeting for review and identification of any opportunities for improvement.

10. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 6 – Action 6.3 Partnering with consumers
- Standard 6 – Action 6.5 Correct identification and procedure matching
- Standard 6 – Action 6.7 Clinical handover
- Standard 6 – Action 6.9 Communication critical information
- Standard 8 – Action 8.4 Recognising acute deterioration
- Standard 8 – Action 8.6 Escalating care

11. References

1. Global PROMPT, Practical Obstetric Multi-professional Training
2. Myles' textbook for Midwives, Sixteenth Edition, 2014 p.477-478 Royal College of Obstetricians and Gynaecologists RCOG Green Top Guideline No 50 Umbilical Cord Prolapse 2014 www.rcog.org.uk. November 2014
3. Yamada T, Kataoka S, Takeda M, Kojima T, Yamada T, Morikawa M, et al. Umbilical cord presentation after use of a trans-cervical balloon catheter. J Obstet Gynaecol Res 2013;39:658–62.
4. Critchlow CW, Leet TL, Benedetti TJ, Daling JR. Risk factors and infant outcomes associated with umbilical cord prolapse: a population-based case-control study among births in Washington State. Am J Obstet Gynecol 1994;170:613–8.

12. Glossary

The following definitions are relevant to this policy.

Term	Definition
Umbilical cord presentation	The umbilical cord lies in front of the presenting part, the membranes are intact.
Umbilical cord prolapse	The cord lies in front of the presenting part and the membranes are ruptured
Occult umbilical cord presentation / prolapse	The cord lies trapped beside the presenting part, rather than below it.

13. Policy owner (relating to these procedures)

Enquiries relating to this procedure may be directed to:

Title: Head of Department
 Department: Obstetrics and Gynaecology
 Email: sangeeta.mallabhat@health.wa.gov.au

14. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4.0	20/09/2024	20/09/2027	Scheduled review

15. Authorisation

This procedure has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Tania Strickland – A/Director Clinical Services
Authorisation date	19/09/2024
Published date	20/09/2024

Appendix 1: Cord prolapse / presentation flow chart

